

Higher-order knockout epistasis is generated by dynamics, not molecular wiring, in 28 Boolean gene regulatory networks

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Abstract

Nonlinear dynamics generate pairwise statistical epistasis from molecular architectures that need not contain epistatic terms [1]; whether this extends to higher-order interactions, and with what magnitude and sign, has not been quantified. Knockout phenotypes reflect composed dynamics, not individual gene rules. Across 28 published Boolean network models (7–18 genes), exhaustive 2^n combinatorial knockout sweeps reveal a systematic composition gap between the interaction spectra of individual gene rules and the attractor-level knockout phenotype. A Walsh–Hadamard decomposition shows that dynamics create higher-order interactions in 19 of 25 non-null networks ($|\Delta_{3+}| \geq 0.5$ pp), with gaps from < 1 to $+83$ percentage points of higher-order spectral energy. In five networks, molecular wiring fails to predict knockout epistasis—gene pairs with the strongest local interactions show the weakest global effects. No joint model of standard topological features (feedback loop counts, rule complexity, in-degree distribution) predicts the gap, with leave-one-out cross-validated $R^2 = -0.27$; individual features correlate negatively, most strongly rule nonlinearity ($\rho = -0.655$). Hill-function ODE conversion preserves the gap sign in all 28 networks tested. Composition gaps are absent in size- and degree-matched random networks, ruling out network size and connectivity as sufficient explanations. Separately, three experimental fitness landscapes show higher-order epistasis beyond additive expectations in real organisms, confirming that higher-order structure exists though this tests an additive baseline rather than the local-rule comparison of the Boolean analysis. Creation dominance is specific to complete knockout. Sweeping the perturbation level from full removal to full activation, the median gap falls from $+6.8$ to -1.0 percentage points and the creation/destruction ratio inverts by half-strength knockdown, with 10 of 27 networks changing sign. These results suggest that predicting higher-order knockout epistasis will require models of the composed dynamics rather than molecular interaction data alone.

Keywords: epistasis, Boolean networks, gene regulatory networks, Walsh–Hadamard transform, composition gap, knockout screens

Author summary

Genes rarely act alone: knocking out multiple genes often produces effects that no single knockout predicts. Biologists call this epistasis. The molecular wiring of a gene regulatory network specifies pairwise interactions, but the phenotype depends on the long-run dynamics of the entire network. We ask: does dynamical composition systematically create or destroy higher-order epistasis? Across 28 published Boolean network models, we find that dynamics create higher-order interactions three times more often than they destroy them. The molecular wiring diagram provides some guidance—local interaction strength correlates with knockout epistasis in most networks—but it fails as a reliable predictor: five networks show reversed correlations, and structural features predict the sign of the gap at chance. These results mean that knockout screens probe an epistatic landscape shaped as much by dynamics as by molecular mechanism. Analysis of three experimental fitness landscapes—a protein resistance landscape, yeast biosynthetic knockouts, and a fungal growth-rate landscape—confirms that higher-order epistasis exceeding pairwise structural predictions is present in real organisms.

1 Introduction

Epistasis—the dependence of phenotype on gene combinations rather than individual effects—is among the oldest measurement problems in quantitative genetics [2, 3]. A persistent difficulty is the gap between molecular and statistical epistasis: the interactions encoded in regulatory logic need not predict the interactions observed in knockout experiments [4]. Gjuvsland et al. [1] showed that nonlinear dynamics generically produce statistical epistasis from gene regulatory architectures, regardless of whether the underlying molecular rules contain epistatic terms. Subsequent work confirmed this in synthetic three-gene circuits, where pairwise and triplet epistasis depended on environment and switched sign across conditions [5].

These results establish that a gap exists. They do not quantify it. The gap has no known magnitude, no known sign structure, and no established dependence on network architecture. One reason is computational: exhaustive knockout measurement requires 2^n subset evaluations, which is infeasible at genomic scale. In Boolean gene regulatory networks of modest size ($n \leq 18$), however, exhaustive evaluation is tractable.

We exploit this tractability across 28 published Boolean networks spanning cell-cycle control, apoptosis, differentiation, DNA repair, and developmental patterning. For each network, we compute two complete interaction spectra using the Walsh–Hadamard transform: one from the local update rules and one from the attractor-level value function. The two spectra are compared order-by-order, yielding a signed composition gap that measures how much higher-order structure dynamics create or destroy (Figure 1).

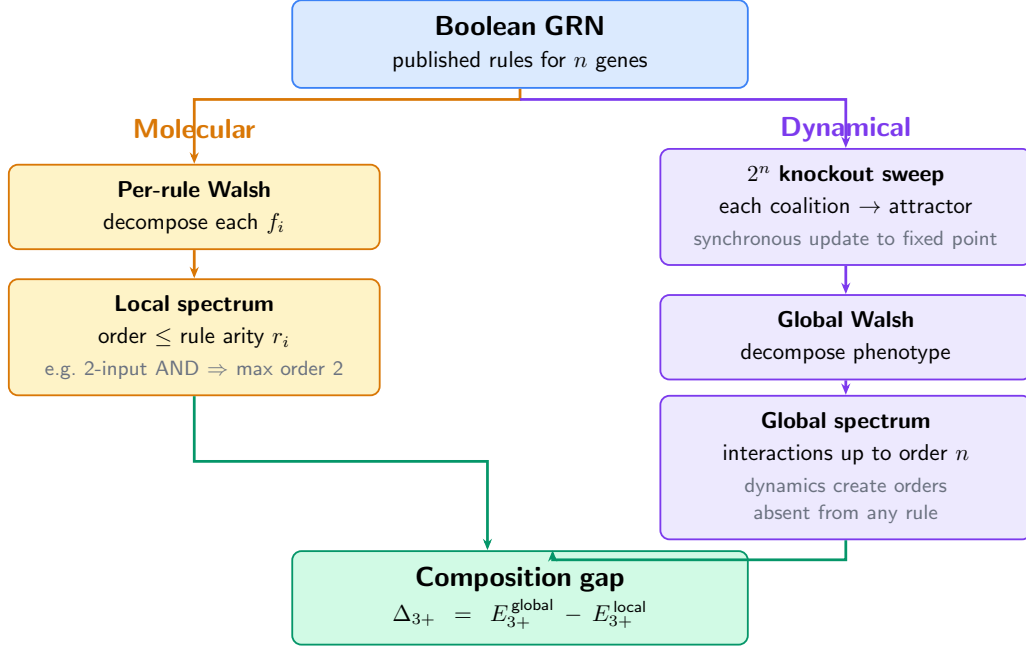


Figure 1: Overview of the composition gap analysis. A published Boolean GRN is analyzed along two paths. The *molecular* path (left) decomposes each gene’s update rule independently via the Walsh–Hadamard transform; interaction order is bounded by rule arity. The *dynamical* path (right) runs an exhaustive 2^n knockout sweep, computes the attractor phenotype for each coalition, and decomposes the resulting value function. The composition gap Δ_{3+} measures how much higher-order spectral energy the composed dynamics create or destroy relative to the local rules.

Three findings emerge. First, creation is the dominant mode: dynamics more often generate higher-order interactions than destroy them (19/25 non-null networks). Second, the molecular wiring diagram is a noisy and sometimes misleading predictor of knockout epistasis—positive on average but negative in five networks. Third, pre-registered structural predictions perform at chance on gap direction (48%), establishing that the composition gap is not reducible to network topology.

2 Methods

2.1 Boolean network models

Twenty-eight published Boolean gene regulatory networks were collected from the Cell Collective repository [6], the PyBoolNet model repository [7], the Biodivine Boolean Models repository, and primary literature. Selection criteria: (i) $n \leq 18$ nodes to permit exhaustive 2^n coalition sweeps within a day of CPU time, (ii) synchronous update semantics used or compatible with the original publication, and (iii) a biologically interpretable output node. The networks span cell-cycle control (mammalian, yeast, *Drosophila*, *Arabidop-*

sis), apoptosis, hematopoietic differentiation, DNA repair, developmental patterning, gene regulation, cancer signaling [8], and signal transduction (Table 1).

To guard against overfitting structural hypotheses to results, the models were analyzed in two batches. Batch 1 (6 networks) was used to develop the analysis pipeline and formulate structural predictions. Batch 2 (21 networks) was analyzed blind: structural predictions were SHA-frozen before any coalition sweep ran. One additional network (Grieco bladder, $n = 18$) was added after the blind protocol and is excluded from blind prediction scoring.

2.2 Local rule Fourier extraction

Each gene k has a Boolean update rule $f_k: \{0, 1\}^{r_k} \rightarrow \{0, 1\}$, where r_k is the number of regulators. We compute the Walsh–Hadamard transform of the truth table:

$$w_T^{(k)} = \frac{1}{2^{r_k}} \sum_{x \in \{0, 1\}^{r_k}} f_k(x) (-1)^{|x \cap T|}, \quad T \subseteq \{1, \dots, r_k\}. \quad (1)$$

A nonzero coefficient $w_T^{(k)}$ at $|T| = m$ indicates a genuine m -way interaction among the regulators of gene k , representation-independent. For each unordered pair $\{i, j\}$ of network genes, the *local pairwise magnitude* is:

$$\ell_{ij} = \sum_{k: i, j \in \text{reg}(k)} |w_{\{i, j\}}^{(k)}|, \quad (2)$$

summing over all genes k whose update rule depends on both i and j . The *local energy spectrum* reports the fraction of total squared-coefficient energy at each interaction order m :

$$E_m^{\text{local}} = \frac{\sum_k \sum_{|T|=m} (w_T^{(k)})^2}{\sum_k \sum_T (w_T^{(k)})^2}. \quad (3)$$

2.3 Coalition sweep and attractor-level value function

For each of 2^n coalitions $S \subseteq \{1, \dots, n\}$, genes not in S are clamped to 0 (knockout). The remaining genes evolve under synchronous Boolean dynamics from $N = 512$ uniformly random initial states until convergence to a fixed point or limit cycle (detected by Floyd’s algorithm, maximum 200 steps). The value function is:

$$v(S) = \frac{1}{N} \sum_{i=1}^N \overline{x_{\text{out}}^{(i)}(S)}, \quad (4)$$

where $\overline{x_{\text{out}}^{(i)}(S)}$ is the time-averaged output-node state for initial condition i under coalition S . The global Walsh-Hadamard transform of v gives the *global interaction spectrum*:

$$w_T^{\text{global}} = \frac{1}{2^n} \sum_{S \subseteq \{1, \dots, n\}} v(S) (-1)^{|S \cap T|}. \quad (5)$$

The global pairwise magnitude is $g_{ij} = |w_{\{i,j\}}^{\text{global}}|$, and the *global energy spectrum* is defined analogously to Eq. 3. The Walsh implementation was verified by exact recovery of analytical coefficients for all 3- and 4-input Boolean functions with known Walsh spectra.

2.4 Composition gap metrics

Pairwise Spearman correlation. For each network, $\rho = \text{Spearman}(\ell_{ij}, g_{ij})$ over all $\binom{n}{2}$ pairs, with bootstrap 95% CIs (10,000 resamples).

Order-3+ energy gap. $\Delta_{3+} = \sum_{m \geq 3} E_m^{\text{global}} - \sum_{m \geq 3} E_m^{\text{local}}$. Positive Δ_{3+} indicates creation; negative indicates destruction. Networks with $|\Delta_{3+}| < 0.5$ pp are classified as null.

Quality gate. The global spectrum is analyzed only if (i) the value function has ≥ 10 unique values, (ii) absolute order-3+ energy exceeds 10^{-4} , and (iii) no single value accounts for $> 90\%$ of coalitions. All 28 networks pass.

2.5 Null model ensembles

Three nested null ensembles test whether the composition gap requires specific regulatory structure. (i) *Kauffman NK*: random networks with the same number of nodes and a uniform in-degree distribution drawn from the same range as the real network. (ii) *Degree-preserving rewired*: networks with the same in-degree sequence as the real network but randomly reassigned regulators and random truth tables. (iii) *Rule-preserving rewired*: the same truth tables as the real network assigned to randomly permuted nodes, preserving each rule's Boolean complexity but changing its regulatory context. For each network and null type, we generated 100 replicates ($n \leq 11$), 30 ($n = 12-14$), or 10 ($n = 15$) and ran the full coalition sweep on each. A quality gate discarded degenerate replicates (fewer than 3 unique Δ_{3+} values or a single value accounting for $> 95\%$ of outcomes).

2.6 Theoretical framework: algebraic and dynamical composition

The composition gap has two mechanistically distinct sources.

Proposition 1 (Acyclic convergence). *If the regulatory graph is acyclic, synchronous dynamics converge to a unique fixed point. The value function reduces to an algebraic composition of local rules, and any composition gap arises from path-wise rule composition alone.*

Proposition 2 (Algebraic creation). *There exist acyclic networks where every local rule has Walsh order ≤ 2 , yet the composed value function has nonzero coefficients at order ≥ 3 .*

Proofs are in Supplement S1. Iterated composition through a depth- d path can raise the Walsh order to $\min(n, d \cdot k)$, where k is the maximum local Walsh order. This algebraic mechanism suffices to explain creation without feedback.

For networks with feedback, a second mechanism emerges: multiple attractors partition the state space, and knockout sets reroute initial conditions across basin boundaries, encoding higher-order interactions. Limit cycles introduce a competing effect: time-averaging the output over a cycle period compresses higher-order coefficients.

Corollary 3 (Decomposition of the composition gap). $\Delta_{3+} = \Delta_{\text{alg}} + \Delta_{\text{dyn}}$, where Δ_{alg} arises from path-wise rule composition (present in all networks) and Δ_{dyn} from attractor basin geometry (zero in DAGs). The sign of Δ_{dyn} depends on whether basin-boundary amplification or cycle-averaging compression dominates.

This predicts that cycling fraction should correlate with Δ_{3+} ; Section 3.3 tests this empirically.

2.7 Hill-function ODE conversion

The Boolean discretization is a modeling choice, not a biological fact. To test whether the composition gap survives under continuous dynamics, we convert Boolean rules to Hill-function ODEs. Each node i evolves as

$$\frac{dx_i}{dt} = \frac{1}{\tau} (\sigma(f_i^{\text{cont}}(\mathbf{x})) - x_i), \quad (6)$$

where f_i^{cont} replaces Boolean operators with smooth approximations (AND $\rightarrow$ product, OR $\rightarrow$ probabilistic OR $a + b - ab$, NOT $\rightarrow 1 - x$) and $\sigma(z) = z^{n_H} / (z^{n_H} + K^{n_H})$ is a Hill sigmoid with cooperativity $n_H = 10$ and half-max $K = 0.5$. Knocked-out genes are clamped to zero via a stiff restoring term.

For each coalition, the ODE system is integrated from 32 random initial conditions using RK45 (SciPy `solve_ivp`, $t_{\text{max}} = 30$, $\text{rtol} = 10^{-4}$). The output is time-averaged over the final third of the trajectory to capture limit-cycle behavior. The coalition value function and Walsh decomposition proceed identically to the Boolean analysis.

The local Walsh spectrum is identical under Boolean and ODE dynamics because the local rules do not change; only the global dynamics differ. The ODE composition gap is therefore $\Delta_{3+}^{\text{ODE}} = E_{3+}^{\text{ODE, global}} - E_{3+}^{\text{local}}$.

2.8 Empirical fitness landscape analysis

To test whether the composition gap generalizes beyond simulation, we apply the Walsh decomposition to three published combinatorially complete fitness datasets: cefotaxime resistance across 32 genotypes from 5 mutations in TEM-1 beta-lactamase [9]; four fitness components across 64 genotypes from 6 biosynthetic knockouts in *S. cerevisiae* [10]; and mycelium growth rate across 186 of 256 genotypes from 8 loci in *Aspergillus niger* [11]. Dataset-specific preprocessing is given in Supplement S4.

For all three, no local interaction model is available—the Weinreich sites make no structural contact at 8 Å, and the Hall and Franke locus-to-pathway mappings are not recoverable from the public data—so the local model is set to additive and the composition gap equals the global order-3+ energy fraction. This is a weaker comparison than the Boolean analysis, which contrasts the measured local rule spectrum against the phenotype spectrum. The empirical arm therefore tests whether higher-order structure exists beyond additivity, not whether composition created it.

3 Results

3.1 Creation is the dominant mode

Table 1 summarizes all 28 networks. Of 25 non-null networks ($|\Delta_{3+}| \geq 0.5$ pp), 19 show creation and 6 show destruction. The median Δ_{3+} is +7.4 pp, and the mean is +19 pp. The gap magnitudes span three orders: from effectively zero (Davidich yeast, −0.4 pp; Remy p53, +0.1 pp) to +83 pp (cell-cycle transcription network). This creation bias indicates that nonlinear dynamics more often generate higher-order epistasis than compress it (Figure 2).

Table 1: Composition gap results for 28 Boolean GRNs, sorted by cycling fraction. ρ : Spearman correlation between local and global pairwise magnitudes. Δ_{3+} : global minus local order-3+ energy (pp). Classification: creation (> 0.5 pp), destruction (< -0.5 pp), or null.

Network	n	ρ	p	Δ_{3+}	Cycling
Arellano root stem	9	0.60	$< 10^{-4}$	+50.6	0.0%
Segment polarity	11	0.21	0.126	+19.8	0.0%
EMT switch	12	0.38	0.002	−3.8	0.1%
Lac operon	13	0.08	0.492	+47.4	0.3%
Cell cycle transcription	9	−0.38	0.021	+83.3	0.4%
Morphogenetic checkpoint	12	0.40	$< 10^{-3}$	+11.9	0.4%

Network	n	ρ	p	Δ_{3+}	Cycling
Calzone cell fate (full)	17	-0.02	0.855	+58.2	0.5%
Mendoza T-helper [†]	13	0.21	0.059	-0.3	1.7%
Saadatpour guard cell	13	0.07	0.572	+51.4	5.7%
Asymmetric cell division	9	-0.10	0.547	+44.1	6.3%
Myeloid progenitors	11	0.51	$< 10^{-4}$	-2.3	6.6%
Tournier apoptosis	12	0.25	0.046	+8.5	7.4%
Irons cardiac	15	0.63	$< 10^{-12}$	+10.9	9.3%
Zañudo T-LGL	12	-0.19	0.120	+33.2	10.3%
Hematopoiesis aging	15	0.63	$< 10^{-12}$	+0.9	10.5%
Calzone cell fate (reduced)	11	-0.07	0.586	+37.0	11.1%
Li budding yeast	11	0.53	$< 10^{-4}$	+6.3	12.5%
Fumia cell cycle	14	0.27	0.011	+52.2	12.5%
Remy p53-Mdm2 [†]	10	0.50	$< 10^{-3}$	+0.1	13.4%
<i>Drosophila</i> cell cycle	14	0.12	0.257	+6.2	14.0%
Lambda phage	7	0.61	0.004	-2.5	16.0%
<i>Arabidopsis</i> cell cycle	14	0.21	0.052	+34.3	16.4%
Davidich yeast [†]	10	0.15	0.312	-0.4	17.7%
Pair-rule module	11	0.66	$< 10^{-7}$	-2.7	17.8%
Faure cell cycle	10	0.41	0.005	-3.0	19.6%
Blood stem cell	11	0.59	$< 10^{-5}$	+4.8	24.5%
Grieco bladder	18	0.10	0.197	+3.8	40.3%
Fanconi anemia	15	0.31	0.002	-5.5	47.5%

[†]Null: $|\Delta_{3+}| < 0.5$ pp.

19 creation, 6 destruction, 3 null out of 28 networks.

3.2 Local-global correlation is positive on average but not universal

The median Spearman ρ between local and global pairwise magnitudes is 0.26 (mean 0.27). Fifteen of 28 networks have significant positive correlations ($p < 0.05$). In these networks, the molecular wiring diagram provides a useful prior for predicting which gene pairs will show knockout epistasis.

Five networks show negative ρ , and one of these (cell-cycle transcription, $\rho = -0.38$, $p = 0.021$) is

statistically significant. In this network, gene pairs with strong local Fourier interactions tend to have *weaker* global knockout epistasis. Negative correlations indicate that dynamical composition can actively invert the interaction structure encoded in local rules.

The remaining eight networks have non-significant positive correlations, where the molecular wiring diagram carries no detectable information about knockout epistasis.

3.3 Limit-cycle prevalence correlates with the composition gap

Corollary 3 predicts that cycling fraction should correlate with Δ_{3+} . A negative association discovered on the 6 batch-1 networks ($\rho = -0.83$) replicates in the same direction on the 21 held-out networks ($\rho = -0.37$, $p = 0.10$), consistent with cycle-averaging compressing higher-order coefficients. Counterexamples exist in both directions, so the association is suggestive rather than definitive (Supplement S3).

3.4 Topological features do not jointly predict the gap

Fitting the composition gap directly on the three feature families named above—feedback loop counts, rule complexity, and in-degree distribution—gives a leave-one-out cross-validated R^2 of -0.27 across the 27 networks, worse than predicting the mean. No joint model of these features generalizes.

Individual features do correlate, and in a consistent direction. The fraction of genes with nonlinear update rules is the strongest single correlate ($\rho = -0.655$, Holm-adjusted $p = 0.003$), followed by positive feedback cycles ($\rho = -0.571$, $p = 0.024$), mean in-degree ($\rho = -0.554$, $p = 0.033$), and mean regulator count ($\rho = -0.536$, $p = 0.043$). Every sign is negative: networks whose local rules are already nonlinear or densely regulated show *smaller* gaps. This is what the gap measures—what composition adds beyond the rules—so rules that already carry higher-order structure leave less to add. Network size alone is uninformative ($\rho = -0.018$).

Structural predictions, SHA-frozen in two batches before any coalition sweep executed, achieved 10/21 (48%) direction accuracy on the 21 held-out networks, indistinguishable from chance; a registered positive-feedback-ratio hypothesis was falsified. Both are consistent with the absence of a usable joint predictor. The registration documents and their freeze records are in the repository. Predicting the gap requires simulating the full composed dynamics. We tested whether more expressive structural features improve prediction. Across all 28 networks, Spearman correlations of Δ_{3+} with mean canalizing depth ($\rho = -0.20$), nested canalizing depth ($\rho = -0.38$), fraction fully canalizing ($\rho = +0.48$), Derrida parameter ($\rho = +0.41$), and mean rule-bias deviation ($\rho = -0.48$) all fail Bonferroni correction (11 tests, smallest adjusted $p = 0.109$). The composition gap is not reducible to canalization structure, sensitivity, or bias of the individual rules.

3.5 The composition gap survives continuous dynamics

To test whether the composition gap is an artifact of Boolean discretization, we converted all 28 networks ($n = 7-18$) to Hill-function ODEs (Section 2.7) and recomputed the coalition sweep under continuous dynamics. All 28 networks preserve the sign of Δ_{3+} (full table in Supplement S2). Among the 24 non-null networks, the median absolute difference between Boolean and ODE Δ_{3+} is 1.4 pp, and magnitudes increase more often than they decrease (17/24), suggesting that smooth Hill-function dynamics amplify basin-boundary effects rather than compress them. The sign persists across 2.6-fold variation in network size ($n = 7-18$) and both creation and destruction modes, establishing the composition gap as a property of regulatory architecture (Figure 4).

3.6 Creation dominance inverts under partial perturbation

The knockout is a complete removal, the most extreme perturbation available. To test whether creation depends on that extremity, we swept the clamp level f applied to perturbed genes across $\{0, 0.25, 0.5, 0.75, 1\}$ in the Hill-function ODE conversion, holding every other parameter fixed. Here $f = 0$ is the published loss-of-function condition and $f = 1$ is full activation.

Clamp level f	Creating	Destroying	Median Δ_{3+}
0.00	17	7	+0.068
0.25	15	12	+0.022
0.50	8	16	-0.012
0.75	7	15	-0.009
1.00	6	15	-0.010

Table 2: Composition gap by perturbation level across 27 networks. Creating and destroying count networks with Δ_{3+} above +0.5 pp and below -0.5 pp respectively; the remainder fall inside that band. Creation dominates only at complete knockout.

The balance inverts between $f = 0.25$ and $f = 0.5$ (Table 2). Ten of 27 networks change the sign of Δ_{3+} , eight of them by $f = 0.25$. Creation is therefore a property of complete removal rather than of composition in general.

The gap persists throughout. No network falls below the 0.5 pp threshold at $f = 0.25$, and only 6 of 27 do so at $f = 1$, so partial perturbation changes the sign of the gap while leaving its magnitude comparable.

Perturbation strength does not drive the higher-order fraction. The normalised order-3+ energy σ_{3+}/σ_{2+} is flat across the sweep, 0.384 at $f = 0$ against 0.387 at $f = 0.75$. Zero-ablation inflates pairwise epistasis by one to two orders of magnitude relative to mean ablation, and that effect does not extend to order 3+.

3.7 Higher-order structure does not sit on the regulatory graph

If dynamics create interactions that the wiring does not contain, then higher-order coefficients should be harder to predict from the regulatory graph than those of a model built to live on that graph. We tested this against the generalised NK model [12], in which each gene j is assigned a random function over its neighbourhood V_j and the phenotype is $\sum_j f_j(x_{V_j})$. For such a model a coefficient is nonzero exactly when its subset lies inside some V_j , which makes it a positive control on the predictor: the predictor recovers 100.0% of GNK order-3+ spectral energy in every network tested.

Applying the same predictor to the attractor phenotype on the same graph, the attractor is less predictable than the GNK model in **24 of 25 networks** (two-sided binomial $p = 1.6 \times 10^{-6}$). The comparison is made within each network, where the graph, the predictor and the fraction of coefficients it marks active are all held fixed and only the phenotype varies. We report the direction rather than the magnitude of the difference: the raw correlations scale with that active fraction ($\rho = 0.975$), so effect sizes are not comparable across networks, while the ordering within a network is unaffected.

Two networks are outside this comparison. *Calzone cell fate* ($n = 17$) exceeds the compute budget for the ten GNK draws per network, and *lambda phage* has every order-3+ subset contained in some neighbourhood, leaving the predictor constant and the correlation undefined. Both are conditions on the input evaluated before any correlation is formed.

3.8 Null models confirm the gap requires specific wiring

Three nested null ensembles (Section 2.5) tested whether network size, connectivity, or rule complexity suffices to produce the observed composition gaps. Mean Δ_{3+} across the 26 testable networks ($n \leq 15$) was +18.6 pp in the real networks, +3.5 pp in Kauffman NK nulls, +3.7 pp in degree-preserving nulls, and +10.3 pp in rule-preserving nulls. The Kauffman and degree-preserving nulls produce less than one-fifth of the real gap magnitude, ruling out network size and in-degree distribution as sufficient explanations. The rule-preserving null recovers roughly half the real gap, indicating that Boolean rule complexity accounts for part of the composition gap. The specific assignment of rules to nodes in the regulatory graph roughly doubles the effect. The composition gap therefore has two necessary ingredients: sufficiently complex local rules *and* their specific wiring arrangement.

3.9 Empirical fitness landscapes show higher-order creation

Table 3 summarizes the Walsh decomposition of three published fitness landscapes. In all three datasets, the composition gap is positive: the global fitness landscape contains higher-order epistasis that no pairwise

Table 3: Composition gap in empirical fitness landscapes. Global order-3+ energy fraction from the Walsh decomposition of measured fitness values. Local model is additive in all cases (no structural contacts for Weinreich; independent pathways for Hall and Franke). Composition gap = global E_{3+} since local $E_{3+} = 0$.

Dataset	Phenotype	n	E_{3+} (%)	E_{1+2} (%)
Weinreich TEM-1	log(MIC)	5	2.9	76.3
Hall yeast	Haploid growth	6	0.1	0.8
Hall yeast	Mating efficiency	6	4.7	9.5
Hall yeast	Sporulation eff.	6	1.2	4.7
Franke <i>A. niger</i>	Growth rate	8	11.4	74.0
Franke <i>A. niger</i>	Growth (excl. imputed)	8	1.1	50.2

structural or pathway interaction predicts.

All three datasets show positive composition gaps. The Hall yeast results are phenotype-dependent: the same six knockouts produce negligible higher-order structure for growth (0.1%) but substantial structure for mating (4.7%), consistent with multi-pathway coupling during conjugation. The Franke landscape’s high E_{3+} (11.4%) drops to 1.1% after excluding lethal genotypes, indicating floor effects at the viability boundary. Per-dataset interpretation is in Supplement S4.

4 Discussion

4.1 Creation dominance

The most robust finding is asymmetry: dynamics create higher-order interactions three times more often than they destroy them (19 vs. 6 non-null networks). This asymmetry has a natural explanation. Local Boolean rules have bounded interaction order (at most r_k -way, where $r_k \leq 6$ in these networks), while attractor-level interactions can involve all n genes through pathway composition. The “space” of possible higher-order interactions grows combinatorially with n but is sparsely occupied by local rules, so dynamics have more room to create new interactions than to suppress existing ones.

Two results bound this account. The asymmetry is specific to complete removal: sweeping the perturbation level inverts the creation/destruction ratio by half-strength knockdown (Section 3.6), so “creation dominates” describes the knockout limit rather than composition in general. Because CRISPR interference produces partial knockdown rather than deletion, the sign of the gap expected in a screen depends on knockdown efficiency.

The interactions dynamics create also do not sit where the regulatory graph would put them. Against a model constructed to live on that graph, the attractor phenotype is the less predictable of the two in 24 of 25 networks (Section 3.7). This is evidence about the regulatory edge list specifically. A structural contact

map carries information an edge list does not, so the result does not transfer to the protein setting, where higher-order epistasis has been reported to localise on pairwise contacts [12].

4.2 Implications for knockout screen design

Where the local-global correlation is positive (Section 3.2), pathway databases provide a useful prior for prioritizing combinatorial experiments. The composition gap adds a more specific guide: in creation networks, dynamics generate interactions absent from any local rule, so a screen guided by molecular interactions alone would miss the strongest epistatic pairs. In destruction networks, the wiring diagram over-predicts interactions that dynamics compress. In the five networks with negative correlations, the wiring diagram actively misleads.

4.3 Relationship to prior work

Gjuvsland et al. [1] demonstrated that statistical epistasis is generic in gene regulatory networks. Our results extend this in three directions. First, the gap is quantified order-by-order rather than detected as present/absent. Second, the gap is signed, with creation as the dominant mode. Third, across 28 networks, no joint model of standard topological features predicts the gap: leave-one-out cross-validated R^2 is -0.27 , worse than predicting the mean. Individual features do correlate, most strongly rule nonlinearity ($\rho = -0.655$), but the relationship does not support prediction, and the higher-order coefficients themselves are poorly localised on the regulatory graph.

The global epistasis literature [13, 14] establishes that nonlinear genotype-phenotype maps reshape interaction spectra. Our local-vs-global comparison operationalizes this reshaping as a quantitative spectral transformation, decomposed by interaction order.

4.4 Generalization and selection bias

Three lines of evidence establish that the composition gap is not an artifact of Boolean discretization: ODE sign preservation in 28/28 networks, phenotype-dependent higher-order structure in the Hall yeast system, and order-3+ energy in the Weinreich and Franke landscapes from organisms with no pairwise structural contacts or shared pathways.

The creation dominance finding (19/6) may partly reflect selection bias. These networks were curated from repositories that over-represent interesting emergent behavior (multistability, complex attractors). If dynamical complexity correlates with creation—a plausible assumption—then the 18/6 ratio may overestimate prevalence in gene regulatory networks generally.

Null model ensembles address this concern directly. Kauffman NK and degree-preserving rewired networks produce mean gaps of +3.5 and +3.7 pp, respectively—less than one-fifth of the +18.6 pp real average (Section 3.8). Size and connectivity alone generate small composition gaps through generic nonlinear dynamics, but the observed magnitudes require the specific regulatory rules and their wiring arrangement.

4.5 Limitations

Three design choices constrain interpretation. First, creation dominance is specific to loss-of-function (clamp-to-0) knockout semantics. Clamp-to-1 sweeps compress gap magnitudes 6-fold and reverse the creation/destruction ratio, because AND and OR gates respond asymmetrically to high vs. low clamping. Second, the primary analysis uses synchronous update. Preliminary asynchronous sweeps on three networks ($n = 10\text{--}12$) preserve the gap sign with reduced magnitude, consistent with update-order stochasticity smoothing basin boundaries. Third, the exhaustive 2^n sweep limits analysis to small networks ($n \leq 18$). Real regulatory modules can involve hundreds of genes; whether the composition gap scales, saturates, or changes character at larger n remains unknown. Additional sensitivity analyses (input-node exclusion, aggregation method) are reported in Supplement S5.

5 Conclusion

Across 28 Boolean gene regulatory networks, we measured the complete Walsh interaction spectrum at both the local rule and attractor levels. Three findings emerge. First, dynamics create higher-order epistasis three times more often than they destroy it: composition is a generically constructive process. Second, the molecular wiring diagram is a useful but unreliable predictor of knockout epistasis—positive on average, negative in five networks, and at chance for predicting the gap direction from structural features alone. Third, limit-cycle prevalence correlates with the composition gap ($\rho = -0.51$, $p = 0.006$), consistent with oscillatory averaging as a compression mechanism, though counterexamples prevent a clean separation. Hill-function ODE conversion preserves the gap sign in 28/28 networks tested, confirming that the effect is a property of regulatory topology rather than Boolean discretization. The same Walsh decomposition applied to three experimental fitness landscapes confirms that higher-order epistasis beyond pairwise structural predictions is present in real organisms, with its magnitude depending on which phenotype is measured.

The complete Walsh spectra, coalition value functions, and pre-registration artifacts are available at [\[URL: Zenodo repository\]](#).

Data availability

All data and code are available at [\[URL: Zenodo DOI\]](#). The repository includes: (i) coalition value functions for all 28 networks as compressed NumPy archives; (ii) wiring diagrams and local Walsh spectra as JSON; (iii) pre-registration documents and the blinding protocol. Analysis protocols and thresholds were committed via SHA-256 hashing before results were examined; freeze records carry the hash, the version pins, and verification instructions, and the recorded hashes verify against the released files. `PROVENANCE.md` records the freeze mechanism for each registration and notes the one set—the blind structural predictions—whose ordering rests on the blinding protocol’s commit date rather than on a hash; (iv) Walsh spectra for the Weinreich, Hall, and Franke empirical fitness landscapes; (v) ODE coalition value functions for all networks with Hill-function conversion; (vi) all analysis and plotting scripts. The Weinreich data originate from Weinreich et al. [9], the Hall data from Hall et al. [10] via the harmslab repository, and the Franke data from Franke et al. [11]. No proprietary data or software is required.

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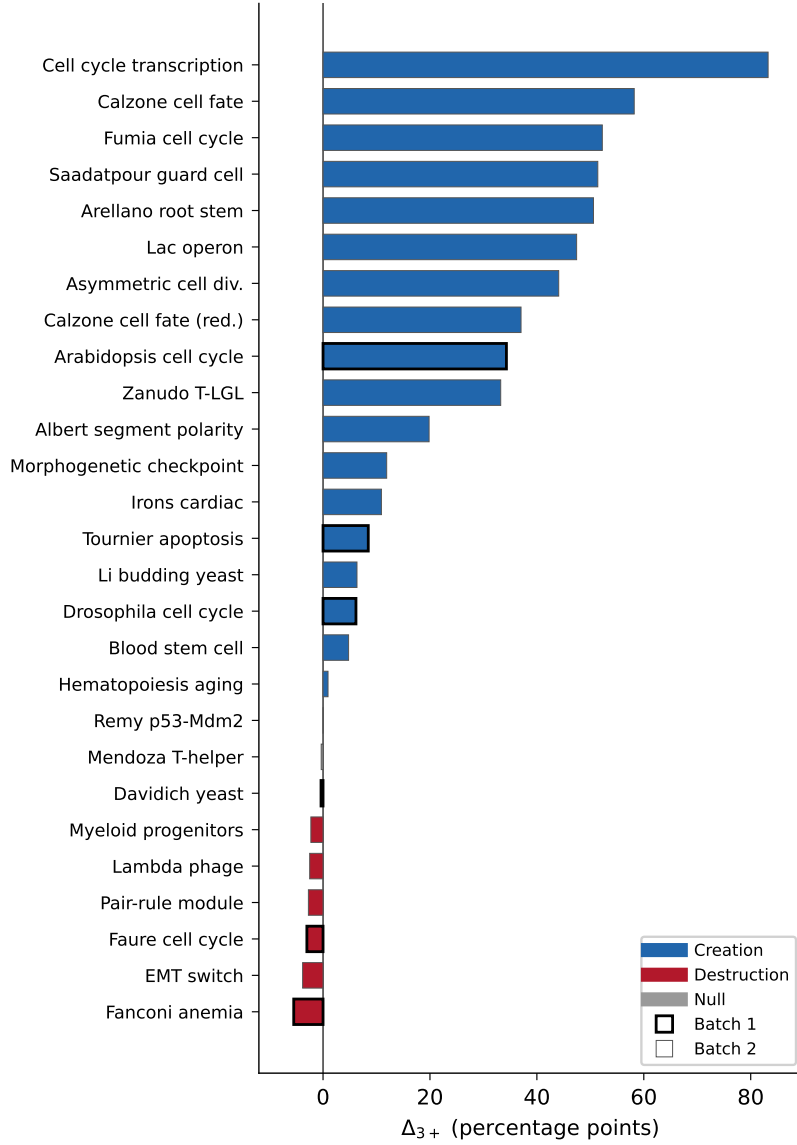


Figure 2: Composition gap (Δ_{3+}) for all 28 networks, ranked by magnitude. Blue: creation (positive gap). Red: destruction (negative gap). Gray: null ($|\Delta_{3+}| < 0.5$ pp). Thick borders indicate batch 1 networks; thin borders indicate batch 2 (held-out). The gap spans three orders of magnitude, from effectively zero to +83 pp.

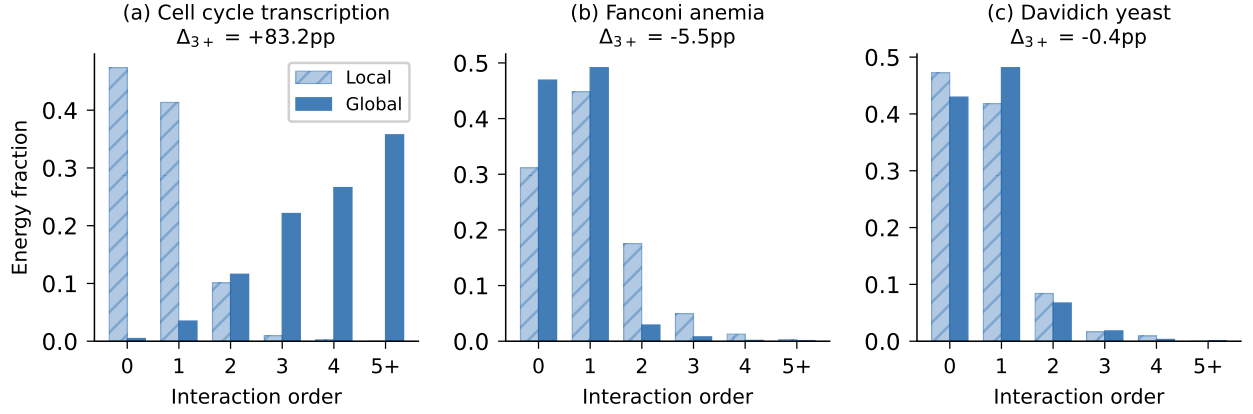


Figure 3: Local (hatched) and global (solid) energy spectra for three representative networks. (a) Cell-cycle transcription ($\Delta_{3+} = +83$ pp): local rules concentrate energy at orders 0–2; dynamics redistribute it into orders 3+. (b) Fanconi anemia ($\Delta_{3+} = -5.5$ pp): dynamics compress higher-order structure. (c) Davidich yeast ($\Delta_{3+} = -0.4$ pp): near-identical spectra (null case).

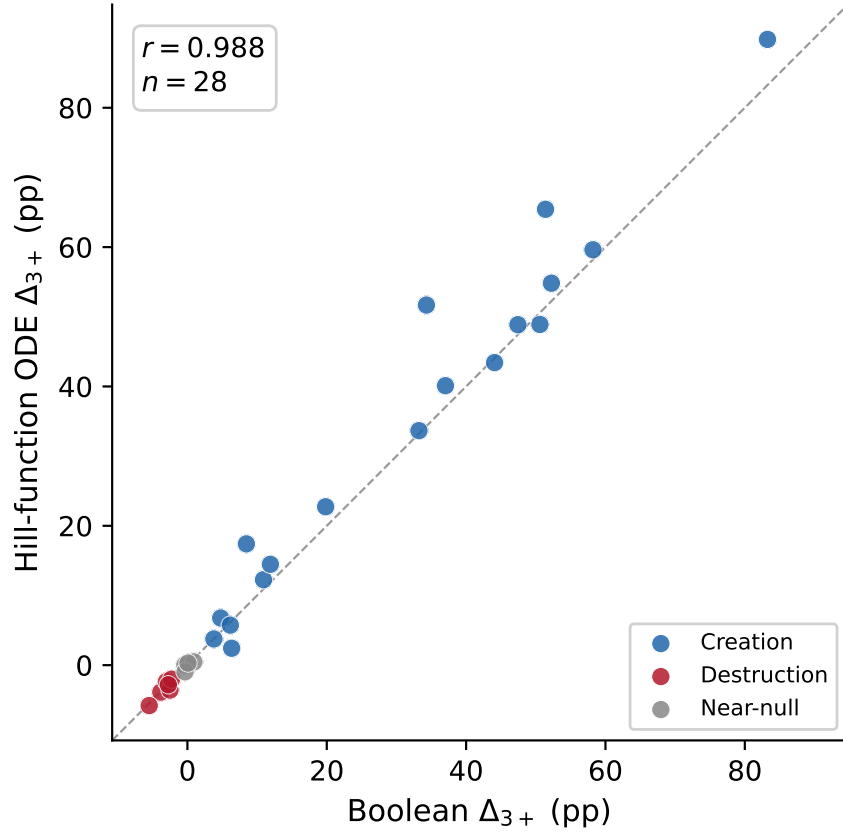


Figure 4: Boolean vs. Hill-function ODE composition gap (Δ_{3+}) across all 28 networks. Each point is one network; dashed line is identity. Blue: creation ($\Delta_{3+} > 0$); red: destruction ($\Delta_{3+} < 0$); gray: near-null. The sign of the composition gap is preserved in every case ($r = 0.988$).

Supplement

S1 Proofs

Proof of Proposition 1. Label the non-input nodes $1, \dots, m$ in topological order. At step t , node i 's update depends only on nodes with indices $< i$. After $d = \text{depth}(\Gamma)$ synchronous steps, every node's value is self-consistent and repeats thereafter. Because each node's steady value is a deterministic function of the source (input-node) values and the clamping configuration, the fixed point is unique. $\square$

Proof of Proposition 2. Consider five nodes x_1, x_2, x_3, h, y with rules $h = x_1 \wedge x_2$ and $y = h \wedge x_3$, a depth-2 DAG. Each rule has Walsh order 2. The composed function $y(S) = (x_1 \wedge x_2) \wedge x_3$ has Walsh expansion (in ± 1 encoding) $y = \frac{1}{8}(1 - z_1 - z_2 - z_3 + z_1 z_2 + z_1 z_3 + z_2 z_3 - z_1 z_2 z_3)$, where $z_i = 1 - 2x_i$. The $z_1 z_2 z_3$ coefficient is nonzero at order 3, absent from every local rule. $\square$

S2 Full ODE comparison table

S3 Cycling fraction details

A negative association between cycling fraction and Δ_{3+} was observed post-hoc on the 6 batch-1 networks ($\rho = -0.83$, $p = 0.04$, but post-hoc so effectively ~ 0.10 two-tailed). The 21 held-out batch-2 networks show the same sign but are not independently significant ($\rho = -0.37$, $p = 0.10$). Pooling all 28 gives $\rho = -0.51$ ($p = 0.006$), but this includes the discovery sample. Among non-null networks, the destruction group has higher median cycling (16.9%) than creation (9.3%), but a Mann-Whitney test is not significant ($U = 79$, $p = 0.09$). Counterexamples: EMT switch has 0.1% cycling but shows destruction (-3.8 pp); blood stem cell has 24.5% cycling but shows creation ($+4.8$ pp). Cycling fraction does not correlate with network size ($\rho = 0.09$, $p = 0.87$).

S4 Empirical landscape details

Datasets and preprocessing

Weinreich TEM-1 beta-lactamase. Weinreich et al. [9] measured cefotaxime minimum inhibitory concentration (MIC) for all 32 genotypes formed by 5 binary mutations in TEM-1 beta-lactamase. MIC values span a $\sim 58,000$ -fold range (0.071–4100 $\mu\text{g/mL}$), so we analyze log-transformed values. The local interaction structure is defined by structural contacts from the crystal structure (PDB 1BTL): two mutation sites

Table 4: Composition gap under Boolean vs. Hill-function ODE dynamics for all 28 networks.

Network	n	$\Delta_{3+}^{\text{Bool}}$	Δ_{3+}^{ODE}	Sign
Cell cycle transcription	9	+83.3	+89.8	preserved
Calzone cell fate (full)	17	+58.2	+59.6	preserved
Arellano root stem	9	+50.6	+48.9	preserved
Saadatpour guard cell	13	+51.4	+65.4	preserved
Fumia cell cycle	14	+52.2	+54.8	preserved
Lac operon	13	+47.4	+48.9	preserved
Asymmetric cell division	9	+44.1	+43.4	preserved
Calzone cell fate (reduced)	11	+37.0	+40.1	preserved
<i>Arabidopsis</i> cell cycle	14	+34.3	+51.7	preserved
Zañudo T-LGL	12	+33.2	+33.7	preserved
Segment polarity	11	+19.8	+22.7	preserved
Morphogenetic checkpoint	12	+11.9	+14.5	preserved
Irons cardiac	15	+10.9	+12.3	preserved
Tournier apoptosis	12	+8.5	+17.4	preserved
Blood stem cell	11	+4.8	+6.8	preserved
Grieco bladder	18	+3.8	+3.8	preserved
Li budding yeast	11	+6.4	+2.4	preserved
<i>Drosophila</i> cell cycle	14	+6.2	+5.8	preserved
Hematopoiesis aging [†]	15	+0.9	+0.5	preserved
Remy p53-Mdm2 [†]	10	+0.1	+0.3	preserved
Fanconi anemia	15	-5.5	-5.8	preserved
EMT switch	12	-3.8	-3.9	preserved
Faure cell cycle	10	-3.0	-2.4	preserved
Pair-rule module	11	-2.7	-2.8	preserved
Lambda phage	7	-2.5	-3.6	preserved
Myeloid progenitors	11	-2.3	-2.0	preserved
Davidich yeast [†]	10	-0.4	+0.04	preserved
Mendoza T-helper [†]	13	-0.3	-1.0	preserved

[†]Near-null under Boolean dynamics ($|\Delta_{3+}| < 1$ pp).

interact locally if their C α -C α distance is ≤ 8 Å. No pairs meet this threshold (closest: E104K-G238S at 12.67 Å), so the local model is purely additive and $E_{3+}^{\text{local}} = 0$.

Hall yeast biosynthetic knockouts. Hall et al. [10] measured four fitness components (haploid growth rate, haploid mating efficiency, diploid growth rate, and diploid sporulation efficiency) for all 64 genotypes formed by 6 biosynthetic gene knockouts in *S. cerevisiae*. Each gene catalyzes a step in a distinct biosynthetic pathway (amino acids, nucleotides, or vitamins). The pathways share upstream precursors from central carbon metabolism (PRPP, TCA cycle intermediates), but the exact locus-to-gene mapping is unavailable from the public data, so the local model is conservatively set to additive ($E_{3+}^{\text{local}} = 0$). All four fitness components are analyzed; haploid growth rate is the primary measure.

Franke *Aspergillus niger* growth rate. Franke et al. [11] report fitness, expressed relative to mycelium growth rate, for 186 of the 256 genotypes formed by 8 binary loci in *Aspergillus niger*. The remaining 70

genotypes were never isolated: 186 segregants were recovered from 2,500 strains tested, and the missing combinations are *assigned* zero fitness on the authors’ statistical argument that they can be treated as non-viable. We analyze the 186 measured genotypes alone and the full 256 with the 70 imputed as zero, since imputed lethality may inflate apparent higher-order interactions through floor effects. The local model is additive ($E_{3+}^{\text{local}} = 0$), as the loci span distinct biosynthetic and developmental pathways.

Per-landscape results

The Weinreich TEM-1 landscape has 2.9% order-3+ energy, dominated by main effects (73.1%) as expected for mutations spread across a 263-residue protein with no pairwise structural contacts at 8 Å. The higher-order structure reflects long-range functional coupling through catalytic efficiency and protein stability—the protein analog of path composition (Proposition 2).

The Hall yeast results are phenotype-dependent. Haploid growth is dominated by the intercept (99.1% order-0), with negligible higher-order structure (0.1%): biosynthetic knockouts have mainly additive effects on vegetative growth. Mating efficiency has 4.7% order-3+ energy, comparable to many Boolean GRNs, consistent with multi-pathway coupling through shared nutrient requirements during conjugation.

The Franke *A. niger* landscape has the highest order-3+ fraction (11.4%) when the 70 unisolated genotypes are imputed as zero. Restricting to the 186 measured genotypes reduces E_{3+} to 1.1%, indicating that the excess arises at the imputed viability boundary rather than among observed strains.

S5 Additional sensitivity analyses

- **Input-node exclusion.** Four networks contain constitutive input nodes ($f(x) = x$). Excluding these from the player set preserves the gap direction in all four cases (median magnitude change 1.0 pp).
- **Aggregation method.** Replacing sum-aggregation with max-aggregation for local pairwise magnitudes (Eq. 2) produces zero sign flips across all 28 networks (median $\Delta\rho = -0.003$).